Hypothesis Testing using Statistical Tests

Learning Objectives

By the end of this session, you will be able to:

- Understand when to use which statistical tests
- Apply and practice the various steps of Hypothesis Testing
- Understand the process of one sample and two sample tests.
- Draw inferences and conclusion.
- Understand the concept of degrees of freedom

What is a Hypothesis and Hypothesis Testing?

- An assumption or educated guess about a population parameter.
- Tested using sample data.

- Hypothesis testing is like a detective game where you :

1. start with an assumption (a hypothesis);
2. then collect evidence (data) to decide whether
3. your assumption is likely to be true or not.

- It’s a way of making decisions or inferences about a population based on a sample of data.

Steps in Hypothesis Testing (Recap)

1. Formulate H_0 and H_1

2. Choose α

3. Select Test

4. Calculate Test Statistic

5. Make Decision

Reference

$$\text{Test statistics} = \frac{\textit{Sample statistic} - \textit{Population Parameter}}{\textit{Standard Error}}$$

Commonly used Critical z values

Significance Level (α)	Two-Tailed Test	One-Tailed Test (Left)	One-Tailed Test (Right)
0.01	±2.58	-2.326	+2.326
0.05	±1.96	**-1.645**	**+1.645**
0.10	**±1.645**	-1.282	+1.282

Statistical Test Computations

Testing of Mean – One sample

Testing of Mean – One sample – Large Sample(>=30) and Population SD is known

Test used: Z test (as the sample size is large and population SD is known)

Test statistic:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Testing of Mean – One sample

Testing of Mean – One sample – Large Sample(>=30) and Population SD is known

Example: Test whether the average height of **18 year old adults in a town** is 158cms with Standard deviation of 2cms. **A sample of 100, 18 year old** was selected and the mean was found to be 159.

Null Hypothesis: There is no significant difference between the sample mean and population mean

Alternate hypothesis: There is significant difference between the sample mean and population mean

Level of Significance: $\alpha = 0.05$

Calculation and Result

Test Statistics: $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

Calculated Value: $Z = \frac{159-158}{2/\sqrt{100}} = 5$ = calculated value

Table value: Z value @ 5% level of significance and two tail test is 1.96 = critical value

Inference: As the calculated value is greater than the table value, Reject Ho. Hence there is significant difference between the sample mean and population mean.

One step ahead: Since the sample mean is greater then the population mean ,

The average height of the population has increased by 1cm. (Inference restricted to this sample study)

Testing of Mean – One sample

Testing of Mean – One sample – **Small Sample(<30)** and Population SD is known

Example: Test whether the average height of 18 year old adult is 158cms with Standard deviation of 2cms. A sample of 25, 18 year old was selected and the mean was found to be 159.

Null Hypothesis: There is no significant difference between the sample mean and population mean

$$H_0: \mu = 158$$

Alternate hypothesis: There is significant difference between the sample mean and population mean

$$H_1: \mu \neq 158$$

Level of Significance: $\alpha = 0.05$

Test Statistics: $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

Calculated Value: $Z = \frac{159-158}{2/\sqrt{25}} = 2.5$

Table value: Z value @ 5% level of significance and two tail test is 1.96

Inference: As the calculated value is greater than the table value, Reject Ho. Hence there is significant difference between the sample mean and population mean.

One step ahead: Since the sample mean is greater then the population mean , The average height of the population has increased by 1cm. (Inference restricted to this sample study)

MCQ

Question: Which test is appropriate for testing a sample mean when the sample size is less than 30 and the population standard deviation is known?

A) F-test

B) T-test

C) Chi-square test

D) Z-test

MCQ

Question: Which test is appropriate for testing a sample mean when the sample size is less than 30 and the population standard deviation is known?

A) F-test

B) T-test

C) Chi-square test

D) Z-test

Answer: D**) Z-test**

Explanation: Even though the sample size is small, if the population standard deviation (SD) is known and the data is approximately normally distributed, a **Z-test** can still be used. However, if the sample size is small and the population SD is unknown, a T-test would be more appropriate.

Testing of Mean – One sample

Testing of Mean – One sample – **Small Sample(<30) and Population SD is unknown**

Test Statistics: t= $\frac{\bar{X}-\mu}{s/\sqrt{n}} \sim$ Student's t distribution with (n-1)df @ 5% level of significance

Where $s = \sqrt{\frac{(X-\bar{X})^2}{n-1}}$

Calculated Value:

Table value: t value with n-1 d.f @ 5% level of significance

Inference: As the calculated value is greater than the table value, Reject Ho. Hence there is significant difference between the sample mean and population mean.

x	X-X-BAR (X – 50)	X-XBAR)2	
80	=80-50 = 30	900	1000/3
30	0	0	333
60	10	100	SQRT(333) = s
30	0	0	

T-table

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768

Degrees of freedom Example

Degrees of Freedom (DoF) represent the number of independent values that can vary in a statistical calculation.

Suppose you have the following 5 exam scores (0 to 100):
x1, x2, x3, x4, and x5 100, 100, 100, 100

You are given that the mean of the scores is 80.

Now suppose the first four scores are allowed to vary freely.

BUT — once 4 of the scores are finalized, the **fifth value is fixed** to maintain the same mean **(80).**

Thus, Degrees of Freedom = n - 1 = 5 - 1 = 4

Testing of Mean – Two Independent sample

Testing of Mean – Two independent sample – Large Sample(>=30) and Population SD is known

Test-statistic:

$$Z = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\bar{x_1}$ and $\bar{x_2}$ are the sample means, σ_1^2 and σ_2^2 are the population variances,

and n_1 and n_2 are the sample sizes.

Two Independent Sample Z-Test

Problem Statement:

- A **researcher** wants to compare the **average daily calorie intake** of **male and female adults** in a city. Two **independent random samples** are taken:

- **Sample 1 (Males):** n1=40, mean X1_bar =2500, population standard deviation σ1=300
- **Sample 2 (Females):** n2=35, mean X2_bar=2300, population standard deviation σ2=250

- At **5% level of significance**, test whether there is a **significant difference** in the mean calorie intake between males and females.

Step 1: define the hypothesis

- We conduct a **two-sample Z-test** because:
  - **Sample sizes are large** ($n1, n2 \geq 30$)
  - **Population standard deviations ($\sigma 1$, $\sigma 2$) are known**

- **Null Hypothesis (H0)**: There is **no significant difference** between the mean calorie intake of males and females. $H0: \mu 1 = \mu 2$

- **Alternative Hypothesis (H1H_1H1)**: There **is a significant difference** between the mean calorie intake of males and females. $H1: \mu 1 \neq \mu 2$

- This is a **two-tailed test**.

Step 2: Compute the Test Statistic

The **Z-test formula** for two independent samples is:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Since H_0 assumes $\mu_1 - \mu_2 = 0$, the formula simplifies to:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Substituting the given values:

$$Z = \frac{2500 - 2300}{\sqrt{\frac{300^2}{40} + \frac{250^2}{35}}}$$

Z comes out to be ?

Step 3: Find the Critical Z-Value

For a **two-tailed test** at **5% significance level (α=0.05)**, we look up the **Z-table**:

Zcritical =±1.96

Step 4: Compare and Make Decision

- **Calculated Z-value = 3.15**
- **Critical Z-value = ±1.96**
- Since Z-calculated > Z critical, **we reject the null hypothesis H0**.

Step 5: Final Inference

- There is **a statistically significant difference** in the mean calorie intake of males and females.

- This suggests that **gender has an influence** on daily calorie consumption.

Session Summary

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- Draw inferences and conclusion.
- Understand the concept of degrees of freedom

Reference Links

- Introduction to Hypothesis Testing
- Frequently asked hypothesis testing interview questions and answers – part 1
- Frequently asked hypothesis testing interview questions and answers – part 2
- Frequently asked hypothesis testing interview questions and answers – part 3
- Sampling methods guide
- How to Collect Data Effectively: Methods, Questionnaire Design, and Data Sources Explained
- Probability Distributions in Python
- Step-by-Step Guide to Two-Sample Z-Test for Large Samples (Population SD Known)